BUILD PLAN · V1

Welded steel workbench

Four square-tube legs, a welded top frame and lower frame, a plywood or steel top and an optional lower shelf. Straight square cuts only: every rail butts onto a leg face.

✓ All 17 checks pass

Make it yours

Change a number and the whole plan, drawings, cut list and checks follow.

Length 1500 mm · Depth 700 mm · Work height 900 mm · Rated load 300 kg · Top Plywood, screwed · Lower shelf yes

OVERALL	MASS	PIECES	WELDS
1500 × 700 × 900 mm	**58 kg**	**16 cut (3 different)**	**28 joints**
length × depth × height	as built, with every part listed	4 plates, 2 sheets	about 4.5 m of weld

RATED LOAD ON THE TOP	POINT LOAD AT A FRONT CORNER	SHOVED SIDEWAYS AT THE TOP	SHOVED FRONT TO BACK AT THE TOP
factor 5.5	**factor 26.8**	**factor 12.2**	**factor 16.1**
sags 1.4 mm	safety factor on yield	safety factor on yield	safety factor on yield

LOADED LOWER SHELF	BUILD TIME	COST	WORK HEIGHT
factor 13.2	**16 h**	**~$136**	**900 mm**
sags 0.4 mm	8 steps	priced lines only; see the materials list	top surface, feet mid-travel

TOP

18 mm plywood

1 Checks

All 17 checks pass at this size.

	Check	Result
✓	**Every joint can be made as drawn** Each end is a mitre, a butt onto a face, a flush cut or a free end; nothing overhangs.	`28 joints`
✓	**Nothing floating** Every member, plate and part touches what holds it.	`all connected`
✓	**No clashes** No member passes through another away from a joint, and no plate through a member it is not notched around.	`none`
✓	**Every piece fits the stock** Cut from standard lengths with a trim and a kerf per cut.	`longest 1420 mm`
✓	**Every piece can be clamped and cut** No piece shorter than 32 mm on its short side.	`shortest 620 mm`
✓	**Rated load on the top: strength** Yield ÷ peak stress with 300 kg spread over the top frame, plus self weight.	`safety factor 5.50 (need 2), worst A (leg)`
✓	**Rated load on the top: deflection** Largest movement of the loaded members.	`1.4 mm (limit 4.9 mm)`
✓	**Point load at a front corner: strength** Yield ÷ peak stress with 150 kg on one front corner, plus self weight.	`safety factor 26.84 (need 2), worst A (leg)`
✓	**Shoved sideways at the top: strength** Yield ÷ peak stress with 400 N pushing the top end-on, plus self weight.	`safety factor 12.24 (need 2), worst C (long rail)`
✓	**Shoved front to back at the top: strength** Yield ÷ peak stress with 400 N pushing the top front edge, plus self weight.	`safety factor 16.12 (need 2), worst A (leg)`
✓	**Loaded lower shelf: strength** Yield ÷ peak stress with 100 kg spread over the shelf, plus self weight.	`safety factor 13.15 (need 2), worst C (long rail)`
✓	**Loaded lower shelf: deflection** Largest movement of the loaded members.	`0.4 mm (limit 5.8 mm)`
✓	**Does not tip over** Free-standing it would tip with 300 N sideways at 900 mm high, so it is fixed down.	`anchored (step 8, an M8 floor anchor through the base of every levelling foot); free-standing factor 0.69`
✓	**Every part has a step** Each piece is cut, fitted or bought in a numbered step.	`8 steps`
✓	**Top supported often enough** Plywood and plate stay flat with supports no more than 600 mm apart.	`487 mm between cross rails`
✓	**Work height** Standing work: about wrist height for a vise, elbow height for fine work.	`900 mm`
✓	**Room above the shelf** At least 300 mm between the shelf and the top frame for boxes and cases.	`604 mm clear`

2 Materials

Stock

Section	Pieces	Cut length	Buy	Mass	Cost
40 × 40 × 2 square tube	16	14.0 m	3 × 6.00 m	33 kg	$72

Plate and sheet

Mark	Part	Size	Qty	Step	Cost
P1	Foot plate	40 × 40 × 6 steel	4	6	$2
P2	Top, plywood	1500 × 700 × 18 plywood	1	8	$19
P3	Lower shelf	1500 × 700 × 18 plywood	1	8	$19

Parts to buy

Part	Spec	Qty	Step	Cost
Hex nut, M12	plain steel, welded to the foot plate	4	6	$1
Levelling foot, M12, bolt-down base	40 mm base with an anchor hole, M12 stud at least 60 mm, 500 kg rating	4	8	$16
Floor anchor, M8	sleeve or wedge anchor, to suit the levelling foot base hole	4	8	$5
Self-drilling screws, countersunk	5.5 × 45 mm, for steel up to 3 mm	24	8	$3

Consumables

- MIG wire, 0.8 mm · 1 kg spool
- Shielding gas, Ar/CO2 · one small cylinder
- Cut-off discs, 1 mm · 5
- Flap discs, 80 grit · 3
- Primer and enamel · 1 L each
- Degreaser and rags

Tools

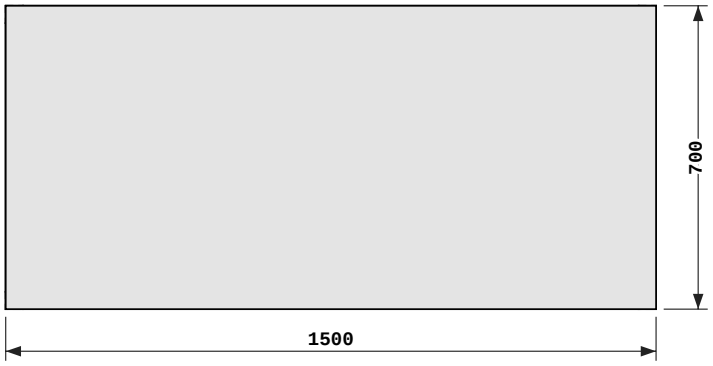
- MIG welder
- Hammer drill and masonry bit for the anchors
- Chop saw or angle grinder with cut-off discs
- Angle grinder with flap disc
- Welding table or flat floor
- Square, tape, scriber, spirit level
- Six or more clamps
- Drill, 13 mm bit, M12 tap

- Circular saw or jigsaw for the plywood
- Impact driver

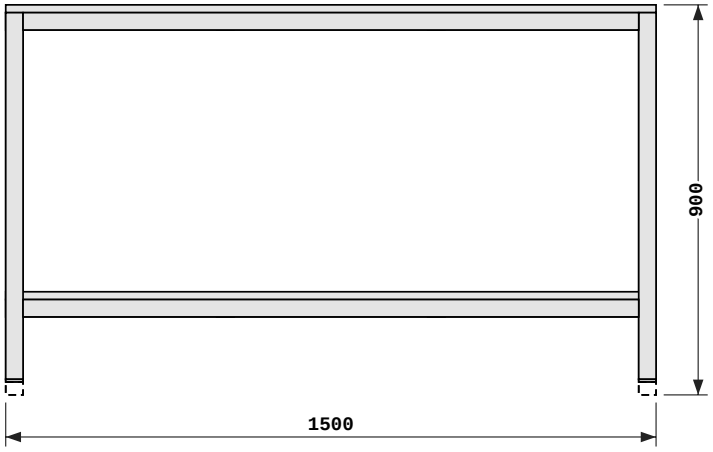
3 Drawings

Overall dimensions. Every number on this page is from the same model as the 3D view.

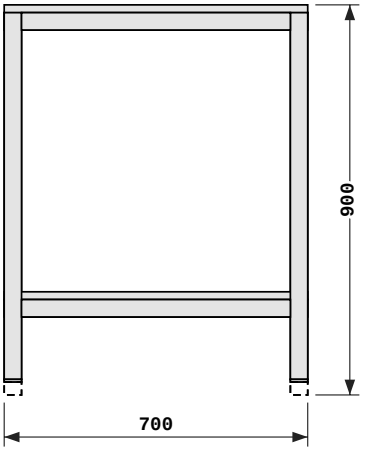
Top



Front



Side (from the right)



4 Legend

Units	millimetres, kilograms, degrees
Datum	x from the left end of the top, y from the front edge, z up from the floor
Piece marks	A, B, C… one letter per different piece; x4 after it means cut four the same. Mirror images share a mark.
Cut marks	From one square end, mark each edge at the two numbers, join the marks around the section, cut on the lines. Angles are for reference.
Edges	named by where they face on the finished build (top-front, bottom-right…)
Joints	mitre = same section, on the bisector · butt onto X = square onto the face of X · flush with X = cut level with X's outer face · then weld
Long point	the overall length of the piece, tip to tip
Dimensions	jig drawings and joint tables give centre lines (℄), where the members' centres cross; the general arrangement gives outside sizes
Tolerance	cut ±1 mm · tacked ±1 mm · finished ±2 mm · diagonals equal within 3

5 Order

Cut everything, tack the end frames, join them, weld out, then feet, paint and top.

Fabricate

1. Cut, deburr, foot plates 3 h
2. Two end frames (tack) 2 h
3. Join the end frames (tack) 2 h
4. Cross rails and shelf rails (tack) 1 h
5. Weld out 3 h
6. Foot plates 1 h

Assemble

8. Feet, top and load test 2 h

Finish

7. Clean and paint 2 h

6 Steps

Ticks save on this device. "Show in 3D" shows the build after that step, the new parts lit.

01 Cut, deburr, foot plates

Fabricate · 3 h

- [ ] Safety glasses, gloves and hearing protection from here on. Use bare or oiled tube, never galvanised.
- [ ] Cut every piece from the cutting plan, square ends; check each cut with a square.
- [ ] Deburr every end and write its piece mark on it in marker.
- [ ] Lay the pieces out by mark and count them against the cut list: 4 legs (A), 12 rails.
- [ ] Cut 4 foot plates 40 × 40 from 6 mm plate and drill a 13 mm hole in the centre of each.

Check Every piece within ±1 mm of its long point on the cut list.

02 Two end frames (tack)

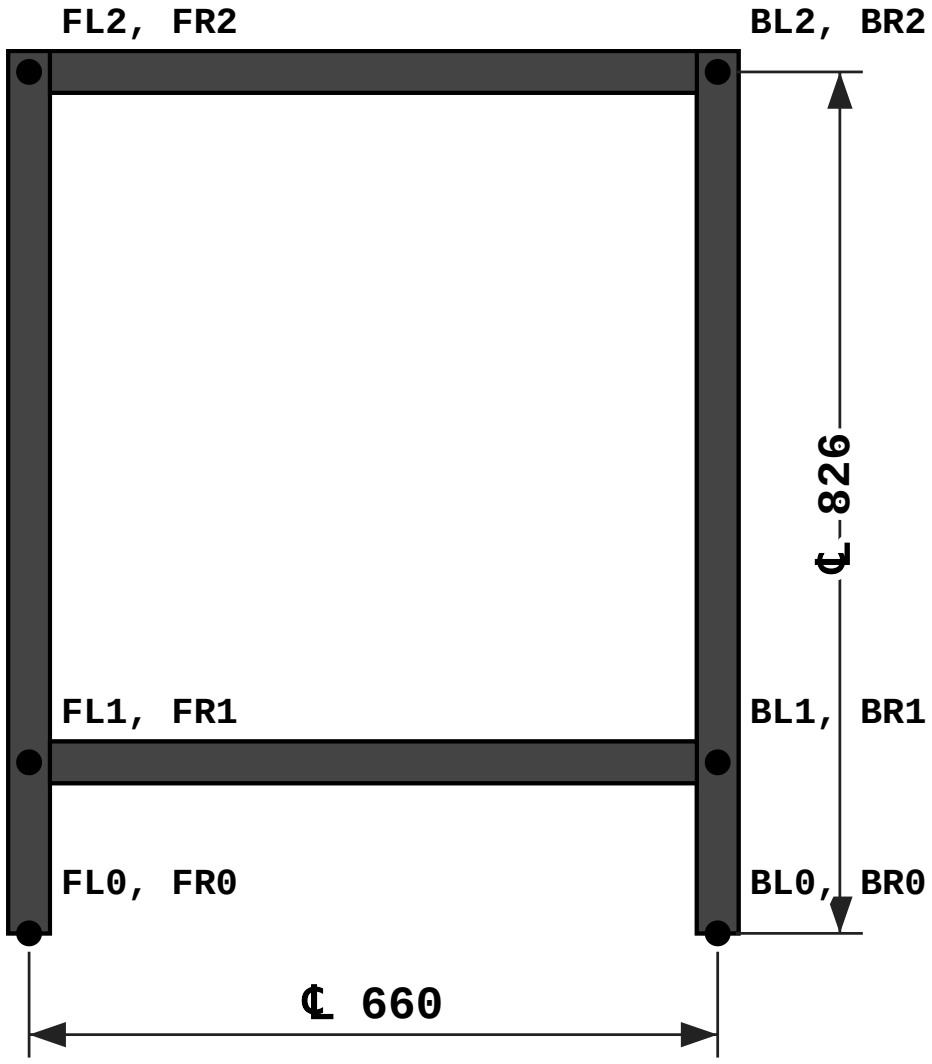
Fabricate · 2 h

- [ ] Lay two legs (A) flat on the welding table, parallel, outer faces against a straight edge.
- [ ] Fit a short rail (B) between them at the top, its top face flush with the leg tops, and a second one lower down with its centre line 164 up from the leg bottoms.
- [ ] Set the outside width to 700 at the top and the bottom, clamp each joint square, tack each joint on two faces.
- [ ] Build the second end frame on top of the first so the two match.

Check Diagonals of each frame equal within 2 mm; both frames the same within ±1 mm; flat on the table at all four corners.

Jig

Step 2: Side (from the right)



Joint positions

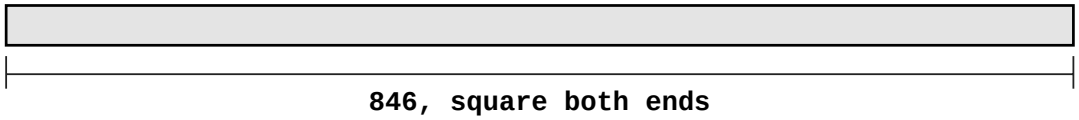
Point	x	y	z
FL0	20	20	36
FR0	1480	20	36
BL0	20	680	36
BR0	1480	680	36
FL1	20	20	200
FR1	1480	20	200
BL1	20	680	200
BR1	1480	680	200
FL2	20	20	862
FR2	1480	20	862
BL2	20	680	862
BR2	1480	680	862

Pieces: A x4, B x4 (of 8)

A x4 — **846 mm**

Leg · SHS 40×40×2 · step 2 — long point

`leg-FL, leg-FR, leg-BL, leg-BR`

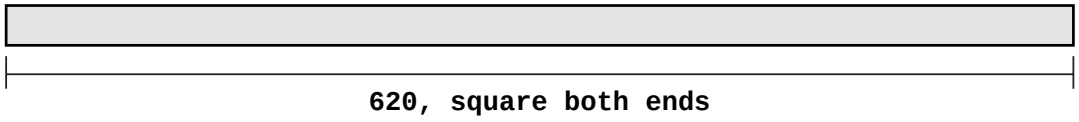


Left end: onto P1 (foot plate) · Right end: flush with B (short rail)

B x8 — **620 mm**

Short rail · SHS 40×40×2 · step 2 — long point

`end-L-top, end-L-low, end-R-top, end-R-low, cross-top-1, cross-top-2, cross-low-1, cross-low-2`



Left end: butt onto A (leg) or C (long rail) · Right end: butt onto A (leg) or C (long rail)

03 Join the end frames (tack)

Fabricate · 2 h

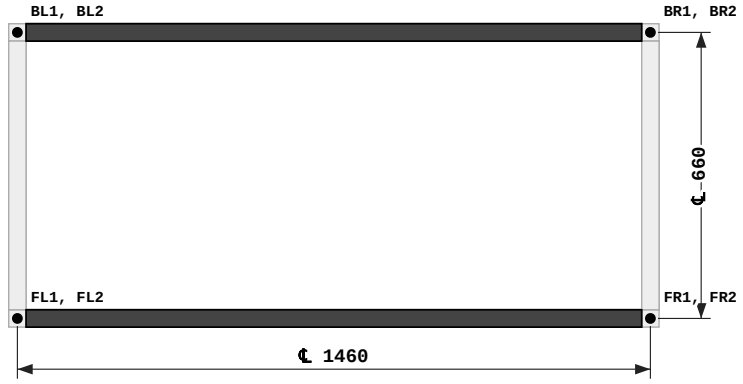
- [ ] Stand both end frames up on the table, top ends up, and brace each with a clamp to a square.
- [ ] Clamp the long rails (C) between the end frames: top rails flush with the leg tops, lower rails at the same height as the lower end rails.
- [ ] Set the outside length to 1500 on the top and the bottom rails.
- [ ] Tack the top long rails first, then the lower ones, one tack per joint.
- [ ] Measure the top diagonals, nudge the frame until they are equal, then tack every joint on two faces.

Check Top diagonals equal within 3 mm; every leg plumb within ±1 mm over its height; the frame sits on all 4 legs without rocking.

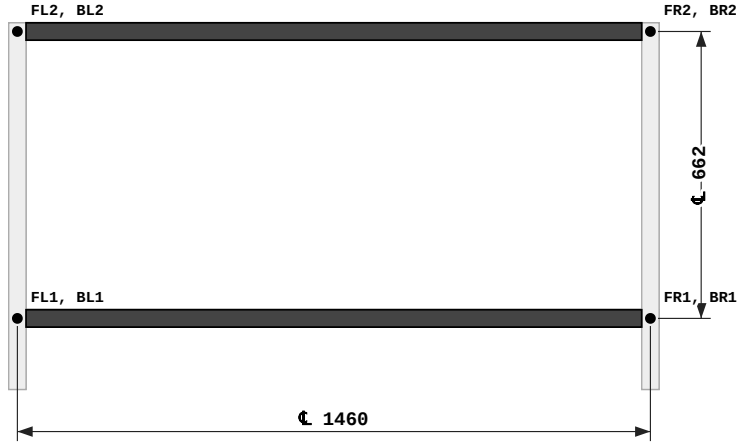
> Tack only: the frame is welded out in step 5, once every rail is in.

Jig

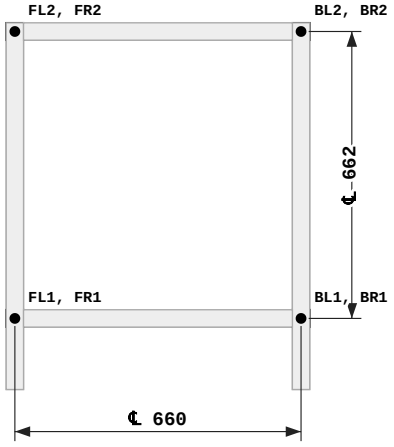
Step 3: Top



Step 3: Front



Step 3: Side (from the right)



Joint positions

Point	x	y	z
FL1	20	20	200
FR1	1480	20	200
BL1	20	680	200
BR1	1480	680	200
FL2	20	20	862
FR2	1480	20	862
BL2	20	680	862
BR2	1480	680	862

Pieces: C x4

C x4

1420 mm

Long rail · SHS 40×40×2 · step 3

long point

`long-F-top, long-B-top, long-F-low, long-B-low`



Left end: butt onto A (leg) · Right end: butt onto A (leg)

04 Cross rails and shelf rails (tack)

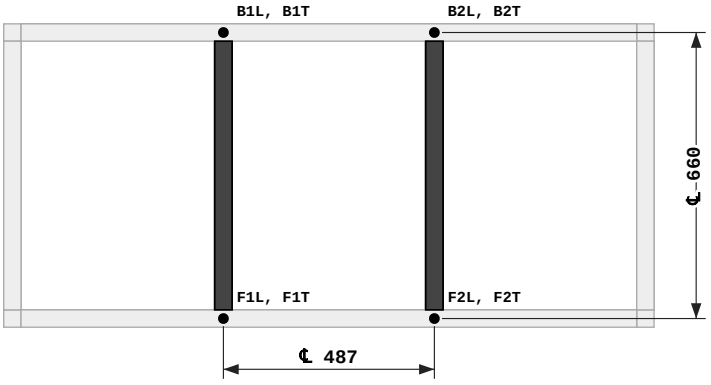
Fabricate · 1 h

- [ ] Mark the cross rail positions on the top long rails, front and back: 507, 993 from the left end to the centre line of each cross rail.
- [ ] Fit a short rail (B) as a cross rail at each mark, between the top long rails, top face flush with them, square to the front rail, and tack both ends.
- [ ] Mark the same positions on the lower long rails and tack a short rail at each as a shelf rail, flush with the top of the lower rails.

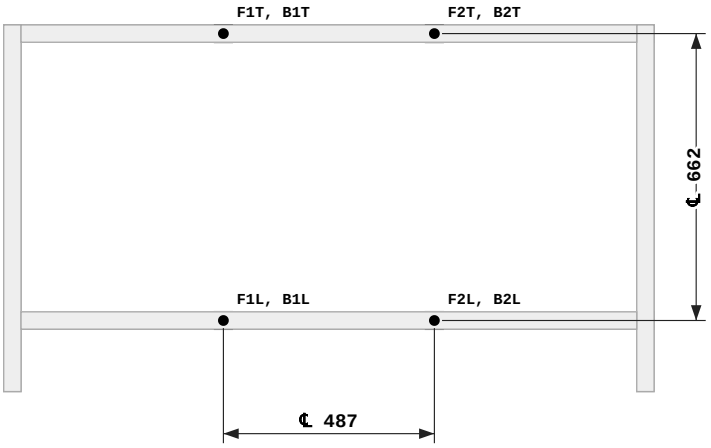
Check Cross rails flush with the top within ±1 mm (lay a straight edge across), and 487 apart centre to centre within ±2 mm.

Jig

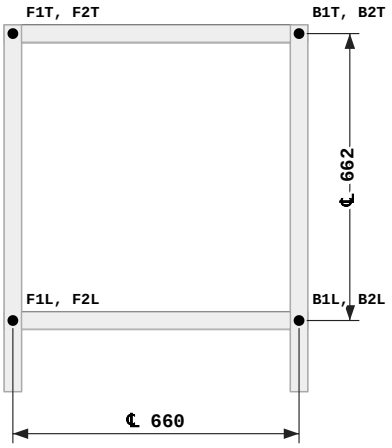
Step 4: Top



Step 4: Front



Step 4: Side (from the right)



Joint positions

Point	x	y	z
F1L	507	20	200
F2L	993	20	200
B1L	507	680	200
B2L	993	680	200
F1T	507	20	862
F2T	993	20	862
B1T	507	680	862
B2T	993	680	862

Pieces: B x4 (of 8)

B: cut sheet in step 2

05 Weld out

Fabricate · 3 h

- [ ] Re-check the top diagonals and the leg plumb before any weld; fix them while it is only tacked.
- [ ] Weld every joint all round in short runs, alternating diagonally opposite corners so the heat does not pull the frame out of square.
- [ ] Weld the top frame joints first, then the lower frame, then the cross rails. Leave it clamped until cold.
- [ ] Grind the welds on the top faces of the top frame flush, so the top sits flat.

Check Top diagonals still equal within 3 mm; a straight edge across the top shows no gap over 1 mm.

> Ventilate: weld fumes from mill scale and paint are toxic.

06 Foot plates

Fabricate · 1 h

- [ ] Weld an M12 nut on the top face of each foot plate, centred over the 13 mm hole: thread a bolt through both to hold it straight while you weld.
- [ ] Lay the bench on its back (two people). Centre a foot plate on the bottom of each leg, nut inside the tube, and weld it all round.
- [ ] Chase each nut with an M12 tap or run a bolt in and out to clear spatter.

Check A bolt screws into every nut by hand, straight.

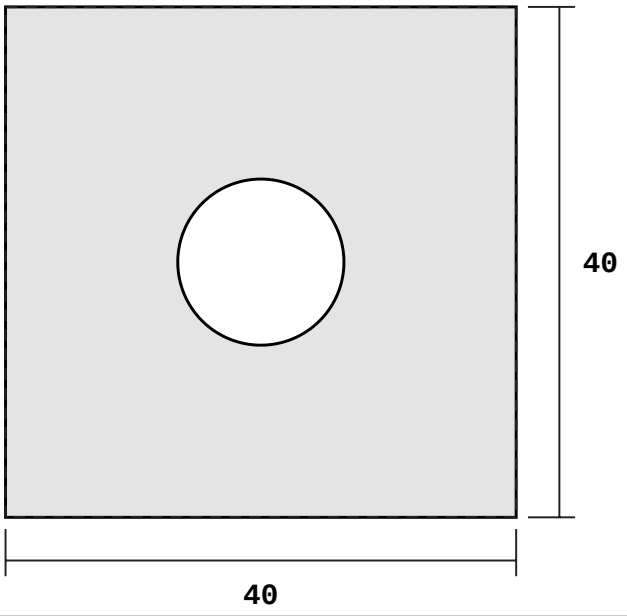
Parts

- Hex nut, M12 ×4 plain steel, welded to the foot plate

Plates

P1 x4

Foot plate · 6 mm steel · step 6



1 hole ø13 mm, centres from the lower-left corner: (20, 20).

Dashed: where a member meets its face (welded on, or under it), from the lower-left corner: (0, 0).

Weld an M12 nut over the hole on the top face first, then weld the plate centred under the leg, all round.

07 Clean and paint

Finish · 2 h

- [ ] Degrease, wire-brush the welds, and sand any mill scale on show.
- [ ] Prime and paint with two coats of enamel; let it dry overnight.

Check No bare steel left except where masked; paint dry to the touch.

08 Feet, top and load test

Assemble · 2 h

- [ ] Screw a levelling foot into each leg and set each to 30 under the foot plate.
- [ ] Cut the top 1500 × 700 from 18 mm plywood, lay it on the frame flush all round.
- [ ] Screw it down into the top frame and cross rails with countersunk self-drilling screws: two per cross rail and per end rail, and two per bay on the front and back rails (12 screws).
- [ ] Cut the shelf 1500 × 700 from 18 mm plywood and cut the notches for the legs as its drawing (step 8) shows.
- [ ] Drop it onto the lower frame and screw it down the same way (12 screws).
- [ ] Stand it in place and level it with the feet: a level on the top, front to back and end to end. Lock each foot with its nut.
- [ ] Mark the anchor hole of every foot base on the floor, move the bench, drill for the M8 anchors, set them, and bolt every foot down (4 anchors).
- [ ] Load it gradually, in steps of about a quarter, up to 300 kg spread over the top, and leave it an hour.

Check Level within ±1 mm over the length; does not move or lift when pushed hard at the top edge; no visible sag or cracked weld after the load test.

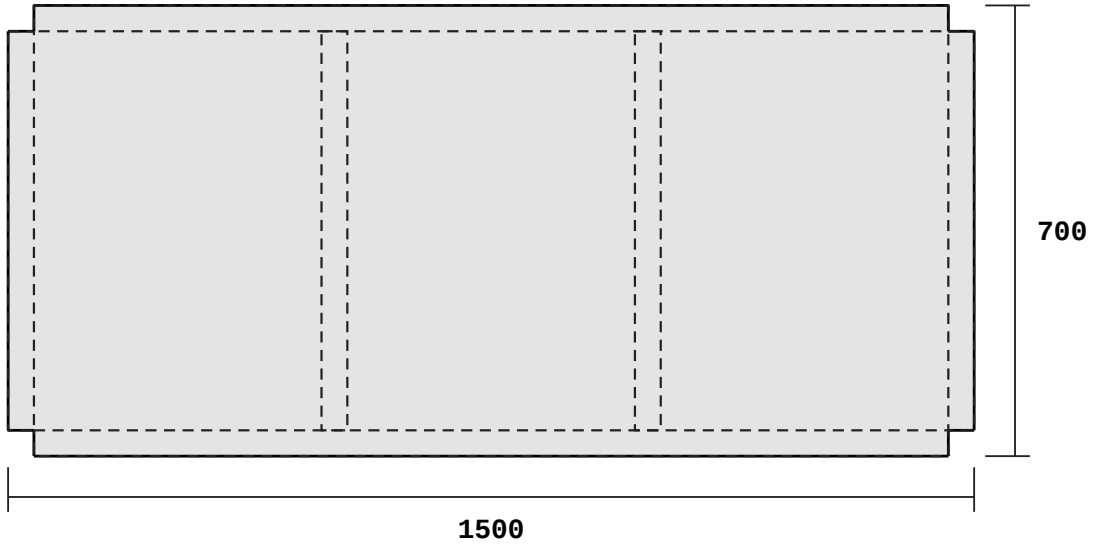
Parts

- Levelling foot, M12, bolt-down base x4 40 mm base with an anchor hole, M12 stud at least 60 mm, 500 kg rating
- Floor anchor, M8 ×4 sleeve or wedge anchor, to suit the levelling foot base hole
- Self-drilling screws, countersunk x24 5.5 × 45 mm, for steel up to 3 mm

Plates

P3 x1

Lower shelf · 18 mm plywood · step 8



4 notches 40 × 40 for the members that pass through, from the lower-left corner: (0, 0) (1460, 0) (0, 660) (1460, 660).

Dashed: where members meet its face (welded on, or under it), from the lower-left corner: (0, 40) (1460, 40) (40, 0) (40, 660) (487, 40) (973, 40).

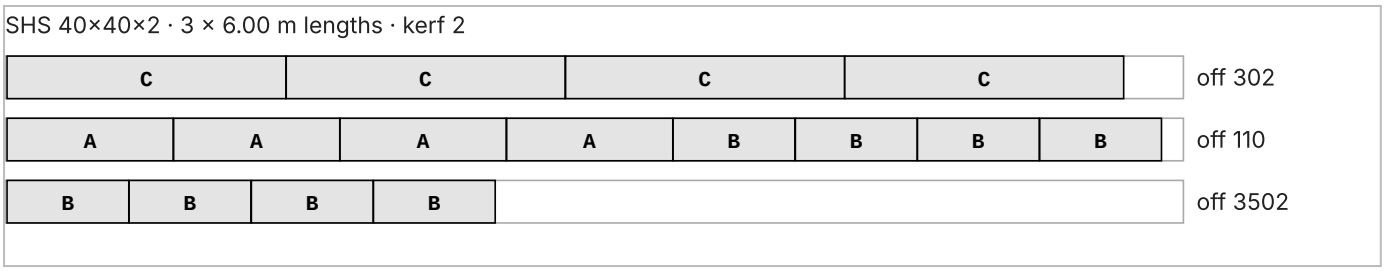
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7 Cut list

Mark	Qty	Section	Long point	Left end	Right end	Step
A Leg	4	SHS 40×40×2	846	square · onto P1 (foot plate)	square · flush with B (short rail)	2
B Short rail	8	SHS 40×40×2	620	square · butt onto A (leg) or C (long rail)	square · butt onto A (leg) or C (long rail)	2
C Long rail	4	SHS 40×40×2	1420	square · butt onto A (leg)	square · butt onto A (leg)	3

Cutting plan

Each bar is one stock length; cut the pieces left to right.



Generated by the blueprint skill. Check every dimension against your own stock before you cut, wear eye and hand protection, and load-test the build gently before trusting it.